

Shuntaro Suzuki

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Ph.D. applicant in computational neuroscience, working on representation learning and foundation models for neural signals, with a focus on decoding the human cognitive processes and advancing brain-computer interfaces.

Education

Keio University
Master of Information and Computer Science

Apr. 2025 – Expected Mar. 2027
Kanagawa, Japan

Keio University
Bachelor of Information and Computer Science

Apr. 2021 – Mar. 2025
Kanagawa, Japan

Publications

Journal Articles

- [1] **S. Suzuki**, S. Nagashima, K. Sugiura. “Cortical-SSM: A Deep State Space Model for Motor Imagery Decoding from EEG Signals”, *Journal of Neural Engineering*, 2026. DOI: 10.1088/1741-2552/ae89e8
- [2] M. Chang, **S. Suzuki**, T. Kurose, T. Ibaraki. “Pretraining alpha rhythm enhancement by neurofeedback facilitates short-term perceptual learning and improves visual acuity by facilitated consolidation”, *Frontiers in Neuroergonomics*, vol. 5, no. 1399578, 2024. DOI: 10.3389/fnrgo.2024.1399578

Conference Proceedings

- [3] **S. Suzuki**^{*}, K. Tokura^{*}, D. Yashima^{*}, K. Amemiya^{*}, K. Sugiura, S. Takamichi. “ELSA: Acoustic Event-Level Semantic Alignment for Fine-Grained Reference-Free Text-to-Audio Evaluation”, *INTERSPEECH*, 2026. (* equal contribution)
- [4] **S. Suzuki**, S. Koyama, S. Hirano, S. Nagashima. “PENGUIN: General Vital Sign Reconstruction from PPG with Flow Matching State Space Model”, *ICASSP*, 2026.
- [5] S. Nagashima, **S. Suzuki**, S. Koyama, S. Hirano. “A Decomposition-Based State Space Model for Multi-variate Time-Series Forecasting”, *ICASSP*, 2026.
- [6] C. Sung^{*}, **S. Suzuki**^{*}, F. Chien^{*}, K. Sugiura, Y. Tsao. “Condition-Invariant fMRI Decoding of Speech Intelligibility with Deep State Space Model”, *ICASSP*, 2026. (* equal contribution)
- [7] D. Yashima, S. Kurita, Y. Oda, **S. Suzuki**, S. Otsuki, K. Sugiura. “ABMAMBA: Multimodal Large Language Model with Aligned Hierarchical Bidirectional Scan for Efficient Video Captioning”, *ICPR*, 2026.

Preprints

- [8] **S. Suzuki**, S. Nagashima, K. Sugiura. “BRAID: A Generalizable EEG Tokenizer via Acquisition-Aware Disentangled Representation Learning”, 2026.

Awards

The 2025 PNPL Competition, NeurIPS, 2025

Aug. 2025 - Sep. 2025

- **Achieved 2nd place** in the Phoneme Classification Task, Standard Track (**USD \$1,000** award).
- Developed a deep learning model for phoneme decoding from SQUID-MEG signals recorded during naturalistic audiobook listening.

Research Experience

Semantic Machine Intelligence Lab, Keio University, Japan

Oct. 2023 – Present

Advisor: Prof. Komei Sugiura

- Developed interpretable deep learning models for motor imagery decoding from ECoG/EEG signals to support communication for patients with ALS, in collaboration with a medical school.

- Constructed a large-scale EEG dataset comprising over 1,000 hours of recordings and conducted acquisition-invariant EEG representation learning across diverse recording protocols and experimental paradigms, including motor imagery and epilepsy.
- Resulting work is currently under review as first-author papers [8], [1].

Suntory Global Innovation Center Limited, Japan

Jan. 2025 – Present

Advisor: Dr. Haruo Mizutania, Dr. Akane Sano, Dr. Nakanishi Masaki

- Developed machine learning pipelines for circadian rhythm analysis using multimodal physiological signals (ECG, PPG, and EEG) acquired from an in-house wearable device.
- Developed an automated artifact detection and removal model for ECG signals using expert annotations.

Neurogica Inc., Japan

Jun. 2025 – Present

- Developed models for reconstructing diverse physiological indicators (e.g., ECG, ABP, Respiratory Rate) from PPG signals, alongside general-purpose architectures for time-series modeling.
- This work resulted in one first-author and one second-author paper accepted to ICASSP 2026 [4], [5].

Biomedical Acoustic Signal Processing Lab, Academia Sinica, Taiwan

Aug. 2025 – Sep. 2025

Advisor: Dr. Yu Tsao

- Developed a deep learning model for speech intelligibility decoding from fMRI signals recorded during speech listening under diverse acoustic conditions, leveraging an extended SSM for efficient fMRI modeling.
- Resulting work accepted at ICASSP 2026 as co-first author [6].

NTT DATA Institute of Management Consulting Inc., Japan

May 2022 – Oct. 2024

Advisor: Dr. Takuya Ibaraki

- Conducted neurofeedback intervention studies using a proprietary wearable EEG device, contributing to experimental design and execution, analysis pipeline development, and manuscript preparation.
- Resulting work published in the peer-reviewed journal Frontiers in Neuroergonomics [2].

Skills

Programming: Python, MATLAB, C/C++, Bash

Machine Learning: Pytorch, Scikit-learn, pandas, Polars, NumPy

Neuroimaging: MNE-Python, Nilearn, NiBabel, EEGLAB

Tools: Git, GitHub/GitLab, Docker, SLURM/HPC

Languages: Japanese (Native), English (TOEFL iBT: 97/120)